Aim:

To implement univariate Linear Regression to fit a straight line using least squares.

Equipment's required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm:

1. Get the independent variable X and dependent variable Y.
2. Calculate the mean of the X -values and the mean of the Y -values.
3. Find the slope m of the line of best fit using the formula.

$$m = \frac{\sum_{i=1}^{n}(x_i - \overline{X})(y_i - \overline{Y})}{\sum_{i=1}^{n}(x_i - \overline{X})^2}$$

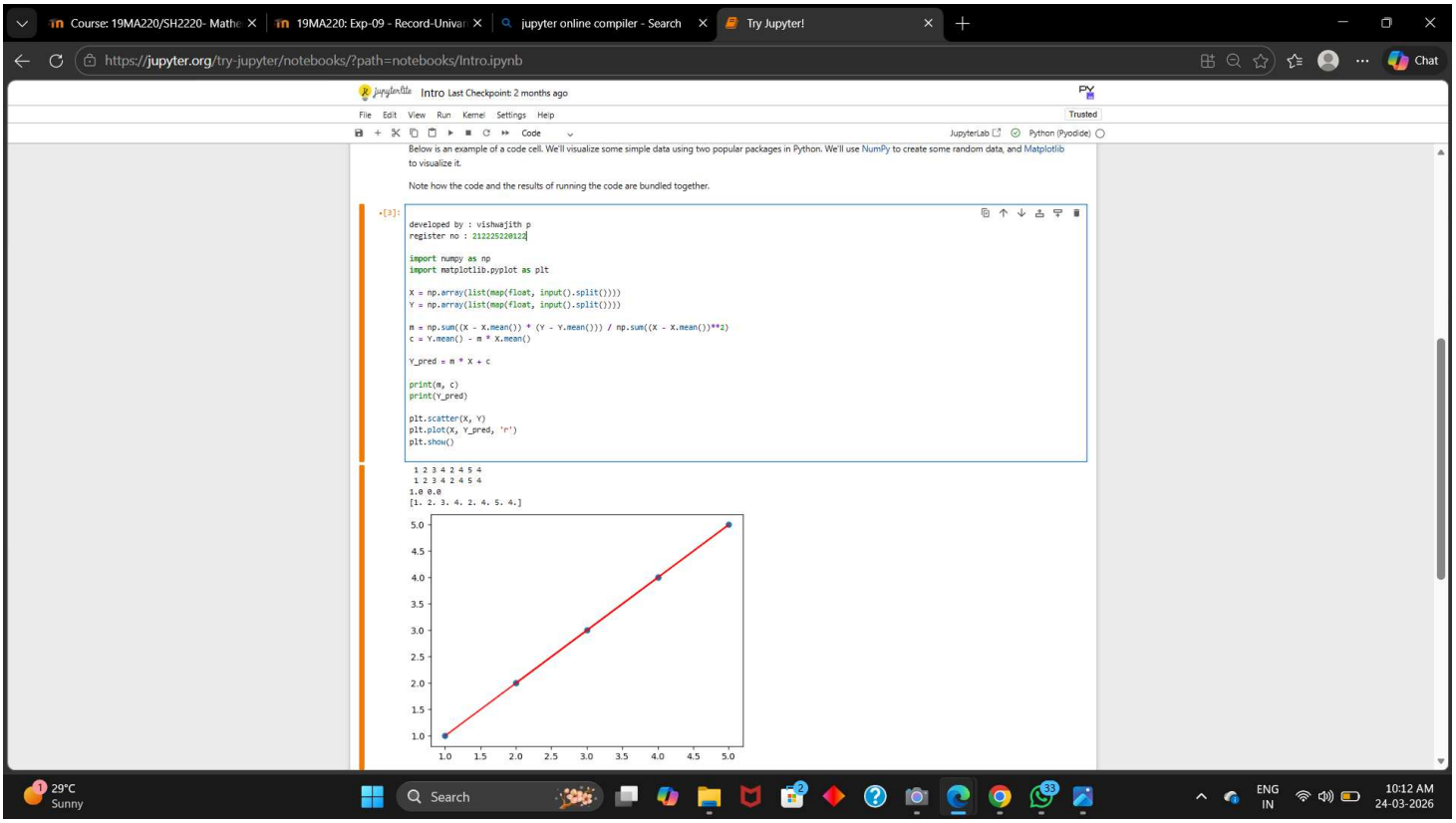
4. Compute the y -intercept of the line by using the formula: $b = \overline{Y} - m\overline{X}$
5. Use the slope m and the y -intercept to form the equation of the line.
6. Obtain the straight line equation Y=mX+b and plot the scatterplot.

Program

```
import numpy as np
import matplotlib.pyplot as plt

X = np.array(list(map(float, input().split())))
Y = np.array(list(map(float, input().split())))

m = np.sum((X - X.mean()) * (Y - Y.mean())) / np.sum((X - X.mean())**2)
c = Y.mean() - m * X.mean()

Y_pred = m * X + c

print(m, c)
```

```
print(Y_pred)

plt.scatter(X, Y)
plt.plot(X, Y_pred, 'r')
plt.show()
```

Output



Result

Thus the univariate Linear Regression was implemented to fit a straight line using least squares.